Open CASCADE Technology
7.4.0

Documentation System

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1 Introduction

This document provides practical guidelines for generation and editing of OCCT user documentation.

2 Prerequisites

You need to have the following software installed to generate the documentation.

Tcl/Tk Version 8.5 or 8.6: https://www.tcl.tk/software/tcltk/download.html

Doxygen Version 1.8.4 or above: http://www.doxygen.nl/download.html

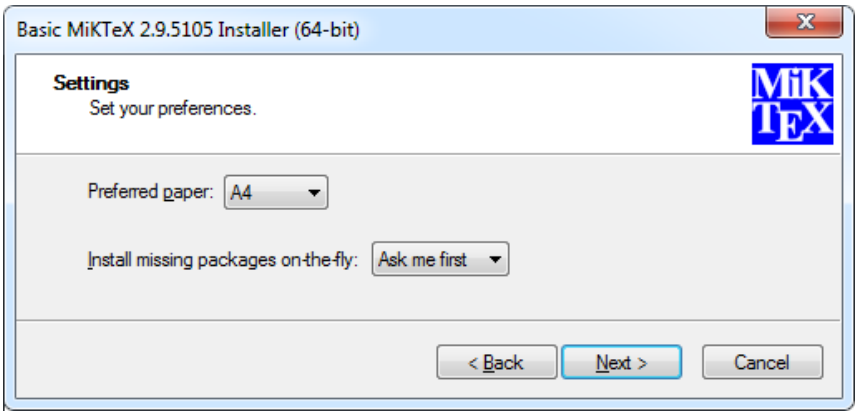
Dot Part of Graphviz software, used by Doxygen for generation of class diagrams in Reference Manual: https://www.graphviz.org/download/

MiKTeX or other package providing **pdflatex** command (only needed for generation of PDF documents): https://miktex.org/download

Inkscape (only needed for generation of PDF documents containing SVG images): http://www.inkscape.org/download

When generating PDF documentation, **pdflatex** and **inkscape** executables should be accessible by PATH variable. You can use *custom.bat* file to add necessary paths to the *PATH* variable.

Note that in the process of PDF generation MiKTeX may need some packages not installed by default. We recommend setting option "Install missing packages on-the-fly" to "Ask me first" (default) during MiKTeX installation:



On the first run of **pdflatex** it will open a dialog window prompting for installation of missing packages. Follow the instructions to proceed (define proxy settings if needed, select a mirror site to download from, etc.).

MathJax is used for rendering math formulas in browser (HTML and CHM outputs): http://www.mathjax.org.

By default MathJAX scripts and fonts work on-line and no installation of MathJAX is necessary if Internet is accessible. If you need to use OCCT documentation while off-line, you can install a local copy of MatJAX, see https://docs.mathjax.org/en/v2.7-latest/start.html#installing-your-own-copy-of-mathjax. See Formulas for more details on inserting mathematical expressions.

3 Documentation Generation

Run command *gendoc* from command prompt (with OCCT directory as current one) to generate OCCT documentation. The synopsis is:

```
gendoc \[-h\] {-refman|-overview} \[-html|-pdf|-chm\] \[-m=<list of modules>|-ug=<list of docs>\] \[-v\] \[-s=
```

Here the options are:

- Choice of documentation to be generated:
  - *-overview*: To generate Overview and User Guides (cannot be used with -refman)
  - *-refman*: To generate class Reference Manual (cannot be used with -overview)
- Choice of output format:
  - *-html*: To generate HTML files (default, cannot be used with -pdf or -chm)
  - *-pdf*: To generate PDF files (cannot be used with -refman, -html, or -chm)
  - *-chm*: To generate CHM files (cannot be used with -html or -pdf)
- Additional options:
  - *-m=<modules_list>*: List of OCCT modules (separated with comma), for generation of Reference Manual
  - *-ug=<docs_list>*: List of MarkDown documents (separated with comma), to use for generation of Overview / User Guides
  - *-mathjax=<path>*: To use local or alternative copy of MathJax
  - *-s=<search_mode>*: Specifies the Search mode of HTML documents; can be: none | local | server | external
  - *-h*: Prints this help message
  - *-v*: Enables more verbose output

Note

- In case of PDF output the utility generates a separate PDF file for each document;
- In case of HTML output the utility generates a common Table of contents containing references to all documents.
- In case of CHM output single CHM file is generated

Examples

To generate the output for a specific document specify the path to the corresponding MarkDown file (paths relative to *dox* sub-folder can be given), for instance:

```
> gendoc -overview -ug=dev_guides/documentation/documentation.md
```

To generate Reference Manual for the whole Open CASCADE Technology library, run:

```
> gendoc -refman
```

To generate Reference Manual for Foundation Classes and Modeling Data modules only, with search option, run:

```
> gendoc -refman -m=FoundationClasses,ModelingData,ModelingAlgorithms -s=local
```

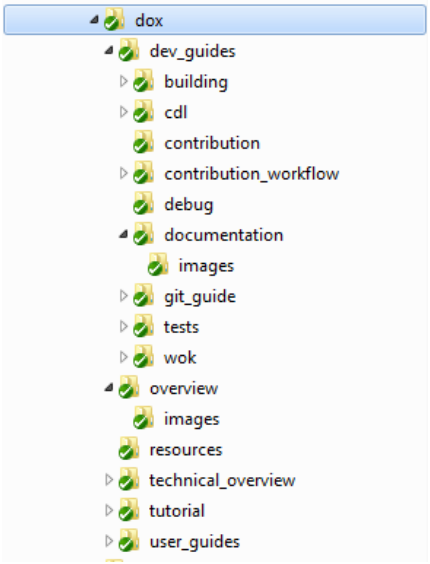
4 Documentation Conventions

This section contains information about file format conventions, directories structure, etc.

4.1 File Format

The format used for documentation is MarkDown with Doxygen extensions. The MarkDown files have a $*.md$ extension and are based on rules described in Appendix 1: Document Syntax section.

4.2 Directory Structure



Each document has its own folder if there are any images used in it. These images are stored in *images* subfolder.

If you want to use the same image for several documents, you can place it in *dox/resources* folder.

Note: To avoid incorrect image display, use a relative path to the image (starting from *dox* folder). For instance:

```
@figure{/dev_guides/documentation/images/documentation_test_image.svg,"",420}
```

The documentation is generated in subfolder *doc* :

- *html* – a directory for generated HTML pages;
- *pdf* – a directory for generated PDF files.

5 Adding a New Document

Place a new document in the folder taking into account its logical position in the documentation hierarchy. For instance, the document *svn.md* about the use of SVN to work with OCCT source code can be placed into */dox/dev↩ _guides/*.

If there are images in the document, it should be placed in its own folder containing a subfolder for images. For instance:

- */dox/dev_guides/svn/* – for *svn.md* file;
- */dox/dev_guides/svn/images/* – for images.

Add a relative path to *svn.md* in file *dox/FILES.txt*. For instance

```
dev_guides/svn/svn.md
```

Note that the order of paths to documents in *FILES.txt* is reproduced in the Table of Contents in the HTML output, thus they need to be placed logically.

Note that you should specify a file tag, not the document name. See Header and hierarchic document structure section for details.

6 Additional Resources

More information about OCCT can be found at http://www.opencascade.com and http://dev.opencascade.org sites.

The information on formula syntax can be found at:
http://en.wikipedia.org/wiki/Help:Displaying_a_formula

More information on MarkDown and Doxygen syntax can be found at:
http://www.stack.nl/~dimitri/doxygen/manual

7 Appendix 1: Document Syntax

A document file in ∗.md format must start with a proper header defining a caption and a unique tag.

```
Documentation System {#dev_guides__documentation}
=====================
```

The document structure is formed by sections that must be defined consistently.

The document can contain plain text, lists, tables, code snippets, images, math, etc. Any specific text elements can be introduced by Markdown language tags or by usual HTML tags.

The table of contents, page numbers (in PDF), and figure numbers (in PDF) are generated automatically.

7.1 Headers and hierarchic document structure

Headers of different levels can be specified with the following tags:

- *@section* – for the first-level headers;
- *@subsection* – for the second level headers;
- *@subsubsection* – for the third level headers.

For example:

```
@section occt_ocaf_1 Basic Concepts
@subsection occt_ocaf_1_1 Applications and Documents
@subsubsection occt_ocaf_1_1_1 The document and the data framework
```

Note that section names can be used for references within the document and in other documents, so it is necessary to use the common prefix indicative of the document name for all section names in the given document. For example, *occt_ocaf* for sections in Open CASCADE Application Framework manual.

The remaining part of section names in most documents consists only of numbers, for example *1_1*. Actually, the hierarchical structure of the output table of contents is not based on these numbers and is generated automatically.

The numbers are only indicative of a section location in the body of the document. However, duplicate section names in a document inevitably cause errors during generation.

If you insert a section in the middle of a big document, do not renumber the document to the end (which is inefficient and error prone), but choose an arbitrary number or letter, not yet used in the document section naming, and base the naming in this section on it.

The section hierarchy is limited to three levels and further levels cannot be presented in the Table of Contents.

However, the fourth and fifth level headers can be tagged with *####* and *#####* correspondingly.

It is also possible to use tags *##* and *###* for second and third level headers if you do not wish to show them in the table of contents or make references to them.

7.2 Plain Text

A plain text is organized in paragraphs, separated by empty lines in MarkDown source. The length of lines is not restricted; it is recommended to put each sentence on a separate line – this is optimal for easier comparison of different versions of the same document.

To insert special symbols, like < , > or \, prepend them with \ character: \<, \>, \\, etc. To emphasize a word or a group of words, wrap the text with one pair of asterisks (∗) or underscores (_) to make it *italic* and two pairs of these symbols to make it **Bold**.

Note that if your emphasized text starts or ends with a special symbol, the asterisks may not work. Use explicit HTML tags and instead.

7.3 Lists

To create a bulleted list, start each line with a hyphen or an asterisk, followed by a space. List items can be nested. This code:

```
* Bullet 1
* Bullet 2
  - Bullet 2a
  - Bullet 2b
* Bullet 3
```

produces this list:

- Bullet 1
- Bullet 2
  - Bullet 2a
  - Bullet 2b
- Bullet 3

To create a numbered list, start each line with number and a period, then a space. Numbered lists can also be nested. Thus this code

```
1. List item 1
   1. Sub-item 1
   2. Sub-item 2
2. List item 2
4. List item 3
```

produces this list:

1. List item 1
   (a) Sub-item 1
   (b) Sub-item 2
2. List item 2
3. List item 3

Note that numbers of list items in the output are generated so they do not necessarily follow the numbering of source items.

In some cases automatic generation adversely restarts the numbering, i.e. you get list items 1. 1. 1. instead of 1. 2. 3. in the output. The use of explicit HTML tags <ol></ol> and <li></li> can help in this case.

Each list item can contain several paragraphs of text; these paragraphs must have the same indentation as text after bullet or number in the numbered list item (otherwise numbering will be broken).

Code blocks can be inserted as paragraphs with additional indentation (4 spaces more). Note that fenced code blocks do not work within numbered lists and their use may cause numeration to be reset.

Example of a complex nested list:

1. List item 1
   Additional paragraph

   ```
   code fragment
   ```

   One more paragraph

(a) Sub-item 1

```
code fragment for sub-item 1
```

(b) Sub-item 2
Paragraph for sub-item 2

Yet one more paragraph for list item 1

2. List item 2

7.4 Tables

A table consists of a header line, a separator line, and at least one row line. Table columns are separated by the pipe (|) character. The following example:

```
First Header  | Second Header
------------- | -------------
Content Cell  | Content Cell
Content Cell  | Content Cell
```

will produce the following table:

First Header	Second Header
Content Cell	Content Cell
Content Cell	Content Cell

Column alignment can be controlled via one or two colons at the header separator line:

```
| Right | Center | Left  |
| ----: | :----: | :---- |
| 10    | 10     | 10    |
| 1000  | 1000   | 1000  |
```

which will looks as follows:

Right	Center	Left
10	10	10
1000	1000	1000

Note that each table row should be contained in one line of text; complex tables can be created using HTML tags.

7.5 Code Blocks

Paragraphs indented with 4 or more spaces are considered as code fragments and rendered using Courier font. Example:

```
This line is indented by 4 spaces and rendered as a code block.
```

A fenced code block does not require indentation, and is defined by a pair of "fence lines". Such line consists of 3 or more tilde (∼) characters on a line. The end of the block should have the same number of tildes. Thus it is strongly advised to use only three or four tildes.

By default the output is the same as for a normal code block. To highlight the code, the developer has to indicate the typical file extension, which corresponds to the programming language, after the opening fence. For highlighting according to the C++ language, for instance, write the following code (the curly braces and dot are optional):

```
~~~{.cpp}
```

```
int func(int a,int b) { return a*b; }
~~~
```

which will produce:

```
int func(int a,int b) { return a*b; }
```

Smaller code blocks can be inserted by wrapping with tags *@code* and *@endcode*.

Verbatim content (same as code but without syntax highlighting) can be inserted by wrapping with tags *@verbatim* and *@endverbatim*.

7.6 Quotes

Text quoted from other sources can be indented using ">" tag. For example:

```
> [Regression in 6.9.0] *IGES - Export of a reversed face leads to wrong data*
```

will produce

> [Regression in 6.9.0] *IGES - Export of a reversed face leads to wrong data*

Note that this tag should prefix each line of the quoted text. Empty lines in the quoted text, if any, should not have trailing spaces after the ">" (lines with trailing spaces will break the quote block).

7.7 References

To insert a reference to a website, it is sufficient to write an URL. For example: http://en.wikipedia.org

To insert a reference to a document or its subsection, use command *@ref* followed by the document or section tag name. For instance,

```
@ref OCCT_DM_SECTION_A
```

will be rendered as Appendix 1: Document Syntax.

Note that links between documents will not work in PDF output if each document is generated independently. Hence it is recommended to add a name of the referenced section after the tag name in the *@ref* command (in quotes): this will guarantee that the reference is recognizable for the reader even if the cross-link is not instantiated. For instance:

```
@ref occt_modat_1 "Geometry Utilities"
```

will be rendered as Geometry Utilities.

7.8 Images

For inserting images into the document use the command *@figure*, as follows:

```
  @figure{/relative/path/to/image/image_file_name.png,"Image caption"}
```

The first argument is a path to the image file, relative to the *dox* folder. The supported formats for images are PNG, JPG, and SVG. The file extension must be lowercase and correspond to the file format. The image file name should have no dots except for the one before extension (names with more than one dot confuse **pdflatex**).

The second argument is optional, it defines the caption for the image to be inserted. The caption argument, if given, should be quoted, even if it is a single word. Captions are included below the image; in PDF output the images with caption are numbered automatically.

Example:

```
@figure{/dev_guides/documentation/images/documentation_test_image.svg,"Test SVG image"}
```

is rendered as:

Figure 1: Test SVG image

We recommend using **Inkscape** for creation and edition of vector graphics. The graphics created in MS Word Draw and some other vector editors can be copy-pasted to Inkscape and saved as SVG images.

Note that the image that will be included in documentation is the whole page of the Inkscape document; use option "Resize page to content" in menu **File ->** **Document properties** of Inkscape to fit page dimensions to the picture (adding margins as necessary).

Note that the *figure* command is an alias to the standard Doxygen command *image* repeated twice: once for HTML and then for Latex output (used for PDF generation). Thus if HTML and PDF outputs should include different images or captions, command "image" can be used:

```
@image html /relative/path/to/image/occ_logo_for_html.png
@image latex /relative/path/to/image/occ_logo_for_pdf.png
```

7.9 Table Of Contents

Use @tableofcontents tag to get the table of contents at the beginning of the document.

Actually, it is not strictly necessary now because TreeView option for HTML is used. The TOC in the PDF document will be generated automatically.

7.10 Formulas

Formulas within MarkDown documents can be defined using LaTeX syntax.

Equations can be written by several ways:

1.Unnumbered displayed formulas that are centered on a separate line. These formulas should be put between @f[and @f] tags. An example:

```
@f[
    |I_2|=\left| \int_{0}^T \psi(t)
            \left\{
                u(a,t)-
                \int_{\gamma(t)}^a
                \frac{d\theta}{k(\theta,t)}
                \int_{a}^\theta c(\xi)u_t(\xi,t)\,d\xi
            \right\} dt
        \right|
@f]
```

gives the following result:

$|I_2| = \left| \int_0^T \psi(t) \left\{ u(a,t) - \int_{\gamma(t)}^{a} \frac{d\theta}{k(\theta,t)} \int_a^{\theta} c(\xi) u_t(\xi,t)\, d\xi \right\} dt \right|$

2.Formulas can also be put between

```
\begin{align}
```

and

```
\end{align}
```

tags.

For example:

```
\begin{align}
\dot{x} & = \sigma(y-x) \\
\dot{y} & = \rho x - y - xz \\
\dot{z} & = -\beta z + xy
\end{align}
```

gives the following result:

$$\dot{x} = \sigma(y-x) \tag{1}$$
$$\dot{y} = \rho x - y - xz \tag{2}$$
$$\dot{z} = -\beta z + xy \tag{3}$$

3.Inline formulas can be specified using this syntax:

```
@f$ \sqrt{3x-1}+(1+x)^2 @f$
```

that leads to the following result: $\sqrt{3x-1}+(1+x)^2$